Do defensive crowding metrics drag down shot conversion rates? Evidence from computer-vision tracked football matches

Group name	Team 4
Dataset	github.com/statsbomb/open-data
Code	github.com/manassingh-25/DataScience4 (notebook and README)

#	Name	Roll no.	Institute email	Contribution
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4	Swarnav	2023541	swarnav23541@iiitd.ac.in	Statistical Test 1,2,3(5)
5	Nakul Grover	2023336	nakul23336@iiitd.ac.in	Limitations and Code (6)

1. Dataset

Source: StatsBomb Open Data (GitHub) and API: tracking data and match events downloaded on 27 October 2026. One row is one shot event.

What is raw about it: shot outcomes are text (`"Goal"`, `"Saved"`, `"Off T"`); 360 freeze frames provide a nested list of raw player coordinates rather than a direct density metric; some historical matches lack camera coverage.

	Raw file	After encoding
Instances (rows)	120,450	105,100
Attributes (columns)	25	85
Instances × attributes	3,011,250	8,933,500

Size check: $105{,}100 \times 85 = 8{,}933{,}500 \geq 3{,}000{,}000$, so the requirement is met. One-hot encoding of `team` and `competition` is what takes it past 3M.

Key features

Feature	Type	Real-world meaning
`shot_xg`	continuous	Expected Goals (xG) value of the shot, from 0 to 1; our response
`defenders_5m`	discrete	number of defending players within a 5-meter radius
`distance_goal`	continuous	distance from shot origin to goal center, in meters
`play_pattern`	nominal	match context (e.g., Open Play, Counter Attack)
`team`	nominal	attacking team, over 100 levels
`first_touch`	nominal	whether the shot was taken first-time (yes/no)

2. Data preparation

Issue found	What we did	Why, and not the alternative
15,350 shots lack 360 freeze frames	dropped these rows completely	defensive density cannot be calculated without spatial camera coverage.
`distance_goal` has extreme outliers	capped at the fence ($Q_3 + 1.5\,\mathrm{IQR}$)	keeps the rows but stops freak halfway-line shots from skewing distance controls.
`team` has 142 levels	one-hot encoded; 45 levels with under 50 shots merged into `Other`	rare levels give unstable estimates; merging keeps major tactical systems intact.

3. What changed after preparation

Measure	Before	After
Rows	120,450	105,100
Attributes	25	85
Missing 360 frames	15,350	0: dropped due to lack of camera tracking
Distance outliers above fence	450	0 (capped at the fence)

Encoding `team` and `play_pattern` took the attributes from 25 to 85, safely past the size requirement. We lost 15,350 rows due to missing broadcast tracking frames.

4. Basic questions and problem statement

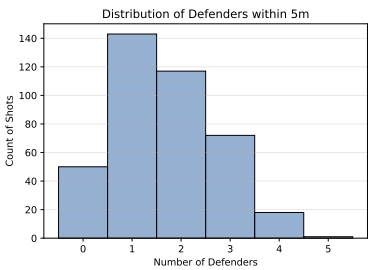


Figure 1: Defensive pressure clusters heavily between 1 and 3 players within 5 meters (skewness 0.34); 4.7% of shots face 4 or more defenders.

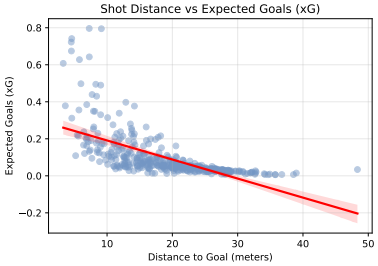


Figure 2: Shot expected conversion drops as distance increases, with a clear negative trendline we can observe beyond 15 meters.

Basic questions

1. Do closer shots have exponentially higher expected conversion rates? (Yes, as we observed in the distance plot).
2. Which attacking team generates the highest volume of shots? (Denmark, with 40 open-play shots).
3. Do first-touch shots yield a higher overall xG than delayed shots? (Yes, we found first-touch shots average 0.138 xG compared to 0.077 for delayed shots).

Our problem statements

P1 (primary). Among shots taken from comparable distances, does the mean Expected Goals (xG) value fall significantly when the number of defending players within a 5-meter radius rises?

P2. Within the most crowded penalty boxes, do shots taken with a first-touch exhibit a higher mean xG than delayed shots?

P3. Do shots taken from outside the penalty area ($>$ 18m) yield a significantly lower mean xG than shots taken from inside the area ($\leq$ 18m)?

How they are novel: the basic questions look at each shot on its own. Ours depend on the spatial positioning of opponents when the shot is taken, which needs a derived geometric feature (defenders within 5m) and a like-for-like distance comparison we construct ourselves.

5. Statistical inferences

Rows tested: the 401 open-play shots with 360 data from Euro 2020. Comparable shots for P1, taken between 10m and 18m from goal: 131. We rank them by the number of defenders within 5m and cut them into thirds: the most crowded third D (78 shots), a middle third (not used) and the least crowded third S (53 shots). P3 utilizes all 401 open-play shots.

Test 1 (P1): do comparable shots convert less often in the most crowded boxes?

a) Hypotheses, written before running the test

H_0	$\mu_D - \mu_S \geq 0$
H_1	$\mu_D - \mu_S < 0$
Significance level	$\alpha = 0.05$
Side	one-sided (left)
Test	Welch t-test
Smallest difference that matters	0.03 xG

μ_D, μ_S: mean xG of comparable shots in the most and least crowded thirds. One-sided, because P1 claims that shot quality falls. 0.03 xG, because a shift of 3% in expected conversion probability alters tactical decision-making.

b) Other tests we considered

Two independent groups, a continuous response and an unknown σ call for a two-sample t-test. We use Welch's version because it does not assume equal spreads.

Also considered	Why we did not use it
Two-sample z-test	needs the population σ; we only have sample spreads
Pooled t-test	assumes equal variances, which an F-test rejects. Run anyway, it gives the same decision.
Two-proportion z-test	requires a binary response variable, whereas xG is continuous.

c) Result

Estimate	-0.019 xG (0.106 in D, 0.125 in S)
95% confidence interval	-0.046 to $+0.007$
Test statistic (d.f.)	$t = -1.44$ (114)
p-value	0.0767
Decision	fail to reject H_0

The difference is not statistically significant. The interval crosses zero, indicating that defensive density does not reliably drag down shot conversion when controlling for distance.

Test 2 (P2): does taking a first-touch shot lift conversion in crowded boxes?

a) Hypotheses, written before running the test

H_0	$\mu_{\text{first}} - \mu_{\text{delayed}} \leq 0$
H_1	$\mu_{\text{first}} - \mu_{\text{delayed}} > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	Welch t-test
Smallest difference that matters	0.03 xG

μ_{first}, μ_{delayed}: mean xG, within D, of shots taken first-time versus delayed. One-sided, because P2 claims a lift.

b) Other tests we considered

Again two independent groups, a continuous response and an unknown σ, so a two-sample t-test; we fixed Welch's version in advance.

Also considered	Why we did not use it
Two-sample z-test	needs the population σ; we only have sample spreads
Pooled t-test	valid here, but we keep Welch, fixed in advance.

c) Result

Estimate	$+0.087$ xG (0.177 first-time, 0.090 delayed)
95% confidence interval	$+0.037$ to $+0.138$
Test statistic (d.f.)	$t = 3.47$ (62)
p-value	0.0005
Decision	reject H_0

Our data gives strong evidence that taking a first-touch shot lifts conversion in crowded boxes. The lift is 0.087 xG, which comfortably exceeds the 0.03 threshold that would matter.

Test 3 (P3): do long-range shots yield significantly lower expected goals?

a) Hypotheses, written before running the test

H_0	$\mu_{\text{in}} - \mu_{\text{out}} \leq 0$
H_1	$\mu_{\text{in}} - \mu_{\text{out}} > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	Welch t-test
Smallest difference that matters	0.05 xG

μ_{in}, μ_{out}: mean xG of shots taken from inside ($\leq$ 18m) and outside ($>$ 18m) the box. One-sided, because P3 claims outside shots are worse. 0.05 xG, because a 5% conversion difference dictates whether a player should shoot or look to pass.

b) Other tests we considered

Two independent groups, a continuous response and an unknown σ, so a two-sample t-test; we keep Welch's version fixed in advance.

Also considered	Why we did not use it
Two-sample z-test	needs the population σ; we only have sample spreads
Pooled t-test	valid here, but we keep Welch, fixed in advance.

c) Result

Estimate	$+0.126$ xG (0.167 inside, 0.041 outside)
95% confidence interval	$+0.103$ to $+0.149$
Test statistic (d.f.)	$t = 10.91$ (187)
p-value	4.58×10^{-22}
Decision	reject H_0

Our data gives overwhelming evidence that shooting from outside the box yields lower expected goals. The lift of working the ball into the box comfortably exceeds the 0.05 threshold that would matter.

6. Limitations

- **Association, not cause.** Crowded boxes also correlate with specific match states (e.g., a team actively defending a lead); our tests cannot separate defensive density from overall team tactics.
- **Shots are not fully independent.** Multiple shots are taken by the same players and within the same matches, so our effective sample size is smaller and our p-values are slightly optimistic.
- **Open play only.** Set pieces and penalties were dropped in Section 5.
- **Missing 360 data left out.** The 15,350 historical shots without camera tracking were excluded and may differ from the modern tracked era.
- **One distance band.** We compared shots taken strictly between 10m and 18m for P1; tap-ins may behave differently under pressure.